

README

Intelligent Hybrid Recommendation System

Project Overview

This project is a Python-based intelligent recommendation system developed in Google Colab. It combines Natural Language Processing (TF-IDF and Cosine Similarity), MBTI personality compatibility, location-based scoring, and an adaptive feedback mechanism to generate personalized user recommendations.

Key Features

- NLP-based profile understanding
- TF-IDF vectorization and Cosine Similarity
- MBTI personality compatibility scoring
- Weighted hybrid recommendation engine
- Adaptive feedback learning
- Top-5 personalized recommendations

Tech Stack

Python, Google Colab, Pandas, NumPy, Scikit-learn, TF-IDF, Cosine Similarity

Project Files

- Major_Project.ipynb – Complete implementation
- users.csv – User profile dataset (50 users)
- match_feedback.csv – Feedback dataset
- Architecture Diagram – Project workflow visualization
- Project Explanation Video – Walkthrough of the implementation

How to Use

1. Open the notebook in Google Colab.
2. Upload the datasets (users.csv and match_feedback.csv).
3. Run all cells sequentially.
4. View the generated compatibility scores and Top-5 recommendations.

About the Project

This project was developed as an academic Machine Learning project to demonstrate hybrid recommendation techniques by combining semantic similarity, personality matching, and adaptive learning.